MANIKANDAN SANTHOSH

SOFTWARE ENGINEER · LLM-INTEGRATED APPLICATIONS

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SUMMARY

Software engineer at AltiusNxt Technologies building full-stack applications that use hosted LLM APIs (Gemini, Perplexity) on business data: sales prospects, product spreadsheets, PDFs and engineering drawings. Built and deployed a TypeScript and React B2B outreach platform, and wrote a CAD drawing analysis agent evaluated on 26 real drawings. Works in Python, TypeScript and JavaScript; integrates external LLM APIs and has not trained or fine-tuned models.

EXPERIENCE

Junior Software Developer
Present

AltiusNxt Technologies · Coimbatore

- Built and deployed AltiusNXT Marketing AI, a B2B outreach platform (TypeScript, Express, React, PostgreSQL): prospect discovery, Gemini-drafted outreach emails with approval gates, sequences and bulk sending
- Deployed the Audit Pipeline, a Gemini-based spreadsheet audit tool with a pilot, review and approve workflow

Trainee Software Developer
Jan 2026 – Jul 2026

AltiusNxt Technologies · Coimbatore

- Worked on CatalogWiz, a product-catalog enrichment and review platform (React, Node.js, MongoDB, Python workers calling Gemini through AWS SQS), including its Shopify and WooCommerce sync

AI Software Developer · Internship
Jul 2025 – Jan 2026

Deep Tech Launchpad · Coimbatore

- Built the DTLP Data Extraction Extension, a Manifest V3 Chrome extension that uses Perplexity or Gemini to extract structured product attributes from web pages, PDFs and Excel lists

SELECTED PROJECTS

AltiusNXT Marketing AI
deployed · sign-in required · private company code

TypeScript · Express · React · PostgreSQL (Prisma) · pg-boss · Gemini API · PM2

- Six pipeline engines: prospect discovery, company enrichment, intent signals, decision-maker discovery, multichannel outreach, engagement tracking
- Bulk email (Excel import, attachments, templates, open tracking); four roles with approval separate from operation; email one-time-code sign-in limited to company domains
- About 74k lines of backend and 32k of React code, 68 database models, about 3,100 test cases (counted); 50 of 58 commits

CAD Drawing Analysis Agent
public source · runs locally

Python · FastAPI · PyMuPDF · Gemini API · NumPy

- Compares five Gemini vision passes with the PDF's own vector geometry and text; labels values HIGH, MEDIUM or LOW and reports conflicts instead of resolving them silently
- Benchmark on 26 real drawings: 90.3% dimension recall, 97.7% value precision (development set, model-labelled ground truth); hole-count accuracy only 28.6%
- 329 passing automated tests; browser workspace with evidence overlays, cited Q&A and Excel/CSV export

CatalogWiz
completed · private company code

React · Node.js · MongoDB · Python · Gemini API · AWS SQS · Shopify API

- Imports product spreadsheets, enriches them with Gemini (attributes, SEO text), runs audit-and-fix, and syncs approved products to Shopify or WooCommerce through a React review interface

DTLP Data Extraction Extension
runs unpacked in Chrome · private repo

JavaScript · Chrome MV3 · Perplexity API · Gemini API · PDF.js

- Extracts user-defined product attributes from pages, PDFs and Excel lists to JSON or CSV; one model per run, no second-model verification, so results need review

TECHNICAL SKILLS

LANGUAGES

Python · TypeScript · JavaScript · C / C++ (coursework)

BACKEND & DATA

Node.js / Express · FastAPI · Flask · REST APIs · PostgreSQL (Prisma) · MongoDB · pg-boss / AWS SQS · pandas · openpyxl

LLM & DOCUMENTS

Gemini API · Perplexity API · Structured JSON prompting · PyMuPDF · pdfplumber · PDF.js · Tesseract.js · Playwright

FRONTEND

React · Vite · Tailwind CSS · Chrome extensions (MV3)

TOOLS

Git / GitHub · PM2 · Vitest / pytest · Shopify API

EDUCATION

KG College of Arts & Science
B.Sc. Computer Technology
2022 – 2025 · Coimbatore · CGPA 7.1

TA Ramalingam Chettiar Hr. Sec. School
Higher Secondary
2020 – 2022

LINKS

Portfolio
manikandan-portfolio-delta.vercel.app — each project page lists evidence and limitations

Public source code
github.com/manikandan-ai17/Pdf-Cad-Drawing-Analizer

Company projects
Private code, not shared publicly; can be walked through in an interview

LANGUAGES

Tamil	Native
English	Professional
Malayalam	Basic